

SHEAR REINFORCEMENT OF WOOD USING FRP MATERIALS

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ABSTRACT: A study of the mechanical behavior of wood members either reinforced or strengthened with fiber-reinforced plastic (FRP) materials in the form of sheets (laminates or fabrics) externally bonded to the shear-critical zones is presented in this paper. The analysis is described first for the (short-term) behavior of shear-reinforced wood members, followed by parametric studies to assess the effect of the type and amount of FRP reinforcement on the response of wood-FRP members. It is shown that the effectiveness of FRP reinforcement can be quite high, and that it is maximized when the fibers are placed in the longitudinal direction and the height of externally bonded sheets is just a little higher than a limiting value, beyond which FRP failure precedes wood failure. In the experimental part of the study, 21 wood beams designed to fail in shear are reinforced with carbon-FRP fabrics at various configurations and area fractions and tested to failure. The agreement between experiments and analysis, in terms of shear capacity, was found to be quite satisfactory.

INTRODUCTION AND BACKGROUND

Wood is a construction material with excellent properties for a variety of applications. Today, improved design methods (e.g., limit state design), preservative treatment techniques, and advanced forms of wood construction (e.g., glulam) enable some contemporary structures, such as small bridges, roofs, and so forth to be at least as permanent and economically competitive as those constructed from other materials. One major disadvantage of wood, however, is its poor strength perpendicular to the grain, sometimes resulting in situations where shear resistance parallel to the grain becomes critical. Curved beams, supports, drilled holes, cut-outs, and possible enhancement of flexural capacity are often situations where shear strength causes trouble.

A number of techniques have been proposed and some of them have actually been applied in practice to increase the shear capacity of wood members. While the majority of these techniques have been developed for the shear reinforcement of new wood members, some of them are applicable to shear strengthening of existing structures as well. Some early studies with shear-reinforced wood employed steel or aluminum plates, which were placed between selected vertical laminations and bonded through the use of epoxy or resorcinol resins (Sliker 1962). Stern and Kumar (1973) developed vertically laminated wood-steel composite beams with one or more steel plates on the sides or between the lumber (flitch beams). Coleman and Hurst (1974) studied the effectiveness of light gauge as a local reinforcement between vertical laminations, in the high horizontal shear stress areas of wood members. For the case of beams with openings, one method of local reinforcement is to glue steel bolts through the stack of laminae, near the opening, and another one is to glue and nail plywood around the opening (e.g., Carling and Johanneson 1988). Problems related to the last two methods include the development of high stresses in the wood due to temperature differences, the development of moisture-induced strains and the negative visual impression effects (in the case of plywood reinforcement).

Fiber-reinforced plastics (FRP) have gained a growing interest as construction materials during the last few decades. These materials are typically made of glass, carbon, or aramid fibers bonded together with a polyester, epoxy, or vinyl ester resin, and possess many desirable characteristics such as high

strength and stiffness, low weight, and immunity to corrosion. Quite a few investigations of FRP as reinforcing materials of wood have been made, aiming at improving durability, esthetics, and mechanical behavior. Most investigators have used fiberglass (GFRP) in various forms parallel to the grain: as face material of wood-core sandwich panels, as external or internal layered reinforcement, and in the form of prestressed strands (e.g., Biblis 1965; Theakston 1965; American 1972; Mitzner 1973; Boehme and Schultz 1974; Saucier and Holman 1975; Rowlands et al. 1986; Moulin et al. 1990; Davalos et al. 1992). Some more recent studies have used advanced composites with carbon (CFRP) or aramid (AFRP) fibers parallel to the grain (Plevris and Triantafillou 1992; Triantafillou and Deskovic 1992; Abdel-Magid et al. 1994; Plevris and Triantafillou 1995), in the form of external reinforcement. Practical applications of CFRP as flexural reinforcement material of wood structures include the strengthening of an historic wood bridge near Sins in Switzerland (Meier et al. 1992), the prestressing of timber beams at the Horrem Community Centre in Germany with GFRP tendons and the prestressing of wood beams in a Japanese tea house with CFRP tendons (Nanni 1993). Studies on the use of composite materials as reinforcement of wood have also included the use of GFRP as reinforcement of timber truss joints, resulting in noticeable increase in their load-bearing capacity (e.g., Chen and Haller 1994).

Studies on the use of FRP as shear reinforcement of wood members have been limited. Blom and Bäcklund (1980) used composites with glass (GFRP) or aramid (AFRP) fibers to reinforce glulam beams in the regions of circular holes. Larsen et al. (1992) increased the shear capacity of curved and cambered glulam beams using GFRP. Finally, Hallström (1995) demonstrated the effectiveness of GFRP as shear strengthening reinforcement of glulam beams in the regions of circular or rectangular holes through both experimental and detailed analytical studies. In the present work, the writer presents a fundamental study of the mechanical behavior of wood either reinforced or strengthened with composite materials in the form of laminates or fabrics bonded to the shear-critical zones. The analysis is described first for the (short-term) behavior of shear-reinforced wood members, followed by parametric studies to assess the effect of the type and amount of FRP reinforcement on the response of wood-FRP members. In the experimental part of the study, the results of an experimental program involving wood beams with shear FRP reinforcement are presented and are found to be in good agreement with the analysis.

ANALYSIS

Shear Capacity

The wood member of Fig. 1(a) is assumed to be reinforced with epoxy-bonded FRP laminates or fabrics in the shear-crit-

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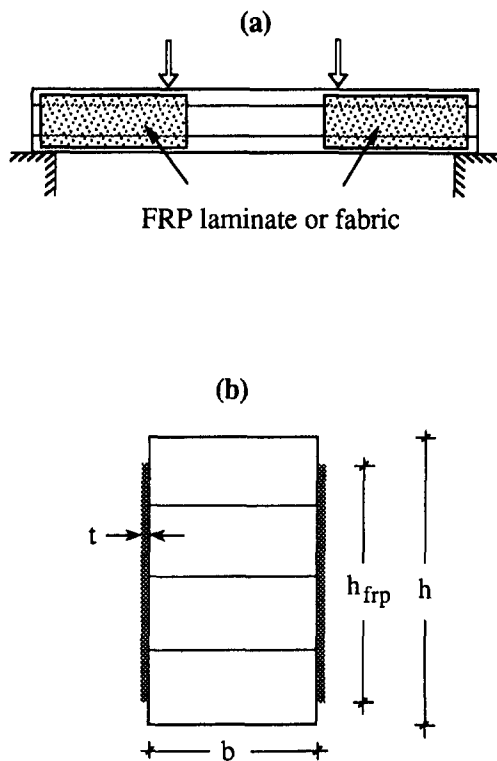


FIG. 1. Diagram of: (a) Wood Beam Reinforced with FRP in Shear; (b) Cross-Sectional Properties

ical zones. The wood section in Fig. 1(b) has a height h and a width b , and the FRP laminates have a thickness t and a height h_{frp} . By transforming the FRP reinforcement to equivalent wood, the maximum shear stress (which occurs in the middle of the section) in the wood is given as

$$\tau_{w,max} = \frac{VQ_{t,max}}{b_t I_t} \quad (1)$$

where V = shear force; and the width b_t , the moment of inertia I_t , and the maximum first moment of area $Q_{t,max}$ of the transformed section are

$$b_t = b + 2t \frac{E_{frp}}{E_w} = b + 2tn \quad (2)$$

$$I_t = \frac{bh^3}{12} + \frac{2nth_{frp}^3}{12} = \frac{bh^3}{12} + \frac{nth_{frp}^3}{6} \quad (3)$$

$$Q_{t,max} = b \frac{h}{2} \frac{h}{4} + 2nt \frac{h_{frp}}{2} \frac{h_{frp}}{4} = \frac{bh^2}{8} + \frac{nth_{frp}^2}{4} \quad (4)$$

In (2)–(4), n = ratio of FRP elastic modulus E_{frp} to wood elastic modulus E_w in the longitudinal direction. Defining the FRP area fraction ρ_{frp} equal to $2th_{frp}/bh$ we can write

$$b_t = b \left(1 + n \frac{h_{frp}}{h} \rho_{frp} \right) \quad (5)$$

$$I_t = \frac{bh^3}{12} \left[1 + n \left(\frac{h_{frp}}{h} \right)^2 \rho_{frp} \right] = I \left[1 + n \left(\frac{h_{frp}}{h} \right)^2 \rho_{frp} \right] \quad (6)$$

$$Q_{t,max} = \frac{bh^2}{8} \left(1 + n \frac{h_{frp}}{h} \rho_{frp} \right) = Q_{max} \left(1 + n \frac{h_{frp}}{h} \rho_{frp} \right) \quad (7)$$

Hence, the maximum shear stress can be written as

$$\tau_{w,max} = \frac{VQ_{max}}{bI} \left\{ \frac{\left(1 + n \frac{h_{frp}}{h} \rho_{frp} \right)}{\left(1 + n \frac{h_{frp}}{h} \rho_{frp} \right) \left[1 + n \left(\frac{h_{frp}}{h} \right)^2 \rho_{frp} \right]} \right\} \quad (8)$$

Failure of the wood in shear will occur when the maximum shear stress $\tau_{w,max}$ becomes equal to the shear strength of wood in horizontal shear, τ_d . Hence, the normalized shear capacity of the FRP-reinforced cross section is obtained as follows:

$$\frac{V}{bh\tau_d} = \frac{2}{3} \left\{ \frac{\left(1 + n \frac{h_{frp}}{h} \rho_{frp} \right) \left[1 + n \left(\frac{h_{frp}}{h} \right)^2 \rho_{frp} \right]}{\left(1 + n \frac{h_{frp}}{h} \rho_{frp} \right)} \right\} = \frac{2}{3} r \quad (9)$$

The factor r in (9) is equal to the ratio of shear capacity of FRP-reinforced section to shear capacity of unreinforced section, and represents the effectiveness of the FRP strengthening reinforcement.

Note that (9) is based on the assumption that no other failure mechanism, such as FRP shear failure or debonding, occurs prior to wood shear failure. The ratio of maximum shear stresses in the FRP and the wood is equal to n , which implies that if the shear strength of FRP, τ_{frp} , is less than $n\tau_d$, then the shear capacity of the cross section is always controlled by the shear strength of plain wood. If such is the case, the normalized shear capacity of (9) is simply equal to 2/3. Debonding of the FRP prior to wood shear failure is not considered in this study. Experience with high-performance adhesives indicates that this failure mechanism is not critical, provided that special care is taken before and during application of the reinforcement. Such care comprises roughening and proper cleaning of the wood and FRP surfaces, uniform spreading of the adhesive, and careful curing until the adhesive hardens.

Parametric Study

The normalized shear capacity predicted by (9) is plotted in Fig. 2 as a function of the FRP area fraction, for various modular ratios n and two ratios of h_{frp}/h . As expected, the increase in shear capacity increases (almost linearly) as ρ_{frp} or n increase and as h_{frp}/h decreases. The last conclusion by no means implies that h_{frp}/h should be very low, because in this case shear failure of wood outside the unreinforced part of the cross section will occur prior to shear failure due to maximum shear stress in the reinforced part. The limiting value of h_{frp}/h , below which such failure is expected to dominate, is given by the condition

$$\frac{Vb[(h - h_{frp})^2/8]}{b_t} = \frac{VQ_{t,max}}{b_t I_t} \quad (10)$$

which leads to the following equation:

$$\left(\frac{h_{frp}}{h} \right)^3 - 2 \left(\frac{h_{frp}}{h} \right)^2 - 2n\rho_{frp} \left(\frac{h_{frp}}{h} \right) + n\rho_{frp} = 0 \quad (11)$$

The solution of (11) is given in Fig. 3 as a function of ρ_{frp} , for various values of n . It can be seen that the limiting value of h_{frp}/h is quite low for practical values of the FRP area fraction, so that wood shear failure outside the reinforced part of the cross section is rather unlikely to occur.

EXPERIMENTAL PROGRAM

To verify the analysis described previously for the shear capacity of wood beams reinforced in shear with FRP laminates, an experimental program was carried out involving

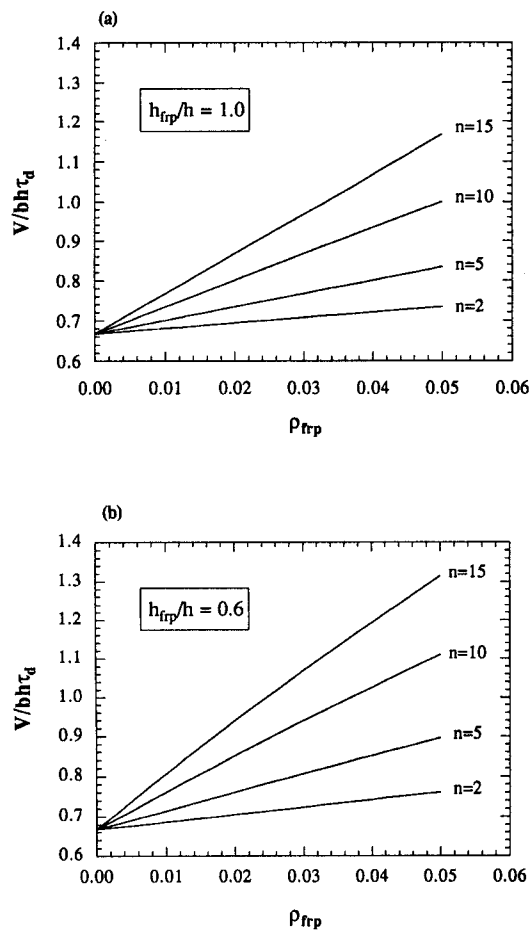


FIG. 2. Shear Capacity As Function of FRP Area Fraction, for Various Modular Ratios

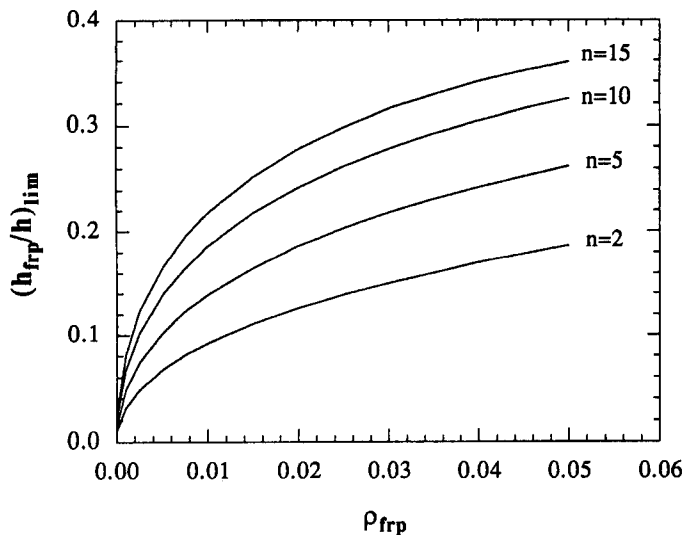


FIG. 3. Limiting Value of h_{fr}/h As Function of FRP Area Fraction

seven different glulam beam designs, with three identical specimens for each design, subjected to four-point bending. Thus, a total of 21 beams, obtained from the same piece of wood, were tested at a total span of 500 mm and a shear span of 110 mm, as shown in Fig. 4. All materials remained at normal room conditions ($20 \pm 2^\circ\text{C}$, $50 \pm 5\%$ RH) throughout the preparation and testing period. The beams had cross sections 65 mm wide and 100 mm deep, except for the shear spans, where the width was intentionally reduced to 25 mm, to ensure

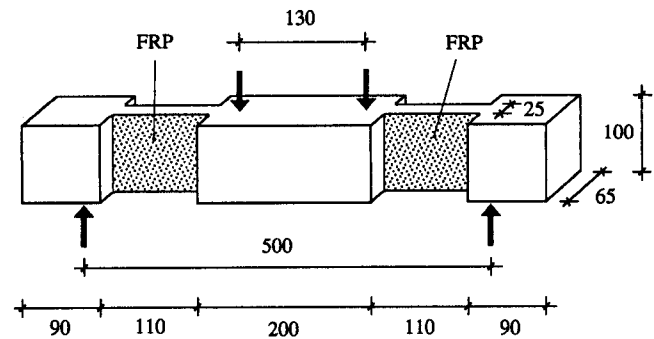


FIG. 4. Geometry of Wood Beams Tested

TABLE 1. Beam Designs Used in Experimental Program

Beam design (1)	CFRP (2)	t (mm) (3)	h_{fr}/h (4)
C	—	—	—
L1/1	one ply at 0°	0.167	1
L2/1	two plies at 0°	0.334	1
LV2/1	two plies at $0^\circ/90^\circ$	0.334	1
V1/1	one ply at 90°	0.167	1
L1/0.6	one ply at 0°	0.167	0.6
L2/0.6	two plies at 0°	0.334	0.6

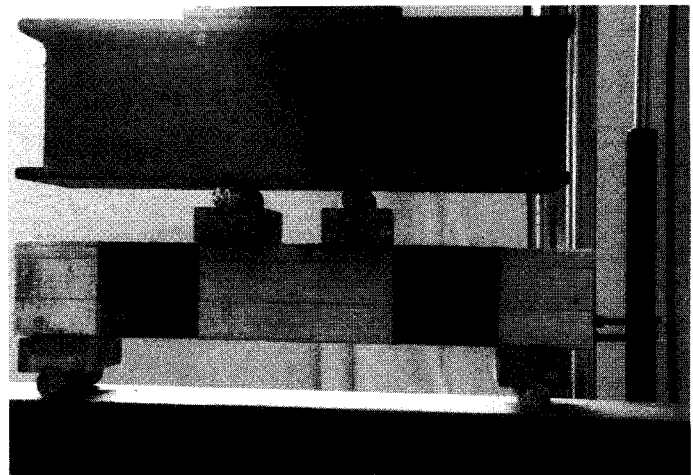


FIG. 5. Experimental Testing Apparatus

that shear failure would be the governing mode. Special efforts were made to obtain uniform and "clean" specimens without knots, to minimize uncertainties associated with variabilities in mechanical properties.

Six out of the seven beam designs were reinforced with epoxy-bonded CFRP fabrics applied to the shear span, and one was left unreinforced, as a control design. A summary of the CFRP area fraction and geometry for each beam design is given in Table 1. It should be emphasized that even though the beams tested were not quite representative of real size members, the relative effect of various CFRP area fractions and orientations (affecting the Young's modulus) on shear capacity could still be demonstrated.

According to the data provided by the CFRP supplier, the fabrics had an elastic modulus of 235 GPa and a tensile strength of 3,300 MPa, both corresponding to a nominal thickness of 0.167 mm. Bonding of the CFRP was achieved through the use of a two-part structural epoxy adhesive (with elastic modulus approximately equal to 5 MPa), following vacuum cleaning of the wood surface. The Young's modulus of wood was measured by recording the load-deflection behavior of three beams identical to the ones described previously (without

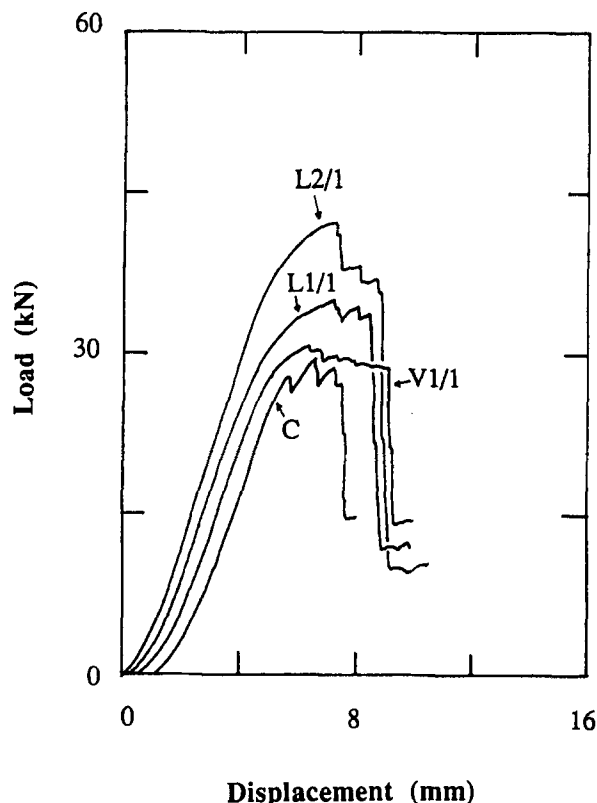


FIG. 6. Typical Load versus Midspan Deflection Diagrams for Wood Beams Reinforced with FRP in Shear

TABLE 2. Theoretical and Experimental Shear Capacity

Beam design (1)	Specimen			Average experimental (5)	Theoretical shear (6)	Difference (percent) (7)
	1 (2)	2 (3)	3 (4)			
C	14.7	13.9	14.9	14.5	14.5	—
L1/1	17.0	18.3	17.5	17.6	18.3	3.8
L2/1	20.2	21.0	21.4	20.7	22.1	6.3
LV2/1	18.0	17.9	17.3	17.7	18.4	3.8
V1/1	15.8	14.8	15.1	15.2	14.6	4.1
L1/0.6	16.2	17.5	17.9	17.2	17.7	2.8
L2/0.6	19.2	17.9	18.9	18.7	20.7	9.7

the width reduction in the shear spans); the beams were loaded in three-point bending over a span of 700 mm.

Loading was applied to the beams approximately one week after the FRP was applied, through a fully programmable servohydraulic MTS testing machine, as shown in Fig. 5, at a rate of 0.05 mm/s. Total load and deflection at midspan were recorded continuously during the tests through the load cell and a linear variable differential transformer (LVDT).

The three-point bending tests on plain wood beams gave an elastic modulus approximately equal to 12 GPa. Also, typical load-deflection curves from the four-point bending tests on plain and shear-reinforced wood beams are illustrated in Fig. 6. In all cases, failure occurred in the shear span, indicating that the dominant failure mechanism was shear. The theoretical and experimentally obtained values for the shear capacity of all beams tested are given in Table 2.

CONCLUSIONS

The concept of reinforcing wood members in shear using externally bonded FRP reinforcement was presented and analyzed in this study. It was shown that: (1) the contribution of

FRP to shear capacity can be predicted based on simple mechanics; (2) the efficiency of FRP reinforcement can be quite high, depending on the FRP configuration and area fraction; (3) for a given increase in shear capacity, the use of FRP is optimized (that is, its area fraction is minimum) when the fibers are placed in the longitudinal direction and the height of externally bonded sheets is just a little higher than the limiting value, beyond which FRP failure proceeds wood failure; and (4) the experimental results for the shear capacity are in good agreement with the analytical predictions (which seem to overestimate the shear failure loads by approximately 3–10%).

In this study it was shown that a little FRP reinforcement can go a long way toward relieving shear stresses in timber structures. This reinforcement method is cost and work efficient and can be applied with minor influence on the esthetics of wood, especially when the FRP materials employed are made of glass fibers in a transparent matrix. Other issues, such as the long-term performance of the FRP-wood system should definitely be addressed, but experiences with FRP-reinforced wood members in bending indicate that no major problems should be expected.

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APPENDIX II. NOTATION

The following symbols are used in this paper:

- b = width of wood cross section;
- b_t = width of transformed cross section;
- E_{frp} = Young's modulus of FRP;
- E_w = Young's modulus of wood;
- h = height of wood cross section;
- h_{frp} = height of FRP sections;
- I = moment of inertia of plain wood section;
- I_t = moment of inertia of transformed section;
- n = modular ratio;
- Q_{max} = maximum first moment of area of plain wood section;
- $Q_{t,max}$ = maximum first moment of area of reinforced section;
- r = reinforcement effectiveness factor;
- t = thickness of FRP reinforcement on each side;
- V = shear force;
- ρ_{frp} = area fraction of FRP reinforcement;
- τ_d = horizontal shear strength of wood;
- τ_{frp} = shear strength of FRP; and
- $\tau_{w,max}$ = maximum shear stress in wood.